Nested Orthogonal Kernel Methods for Longitudinal Distributional Causal Effects

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Abstract

We propose doubly robust kernel tests for longitudinal causal inference based on nested counterfactual mean embeddings. The RKHS-valued efficient influence function decomposes stagewise: each time step induces a distinct inverse problem, and estimation errors propagate forward in time. A single global representer ignores causal ordering, amplifies ill-posedness, and destroy rate guarantees; we propose stagewise Riesz regression in here to respect the structure to estimate the canonical gradient.

Our framework addresses two questions: (Q1) are two longitudinal counterfactual outcome distributions different? and (Q2) where are they different?

1 Problem statement

Data Fix a finite horizon $K \geq 0$. For each unit $i = 1, \ldots, n$, we observe a longitudinal trajectory

$$O_i = (L_{0,i}, A_{0,i}, L_{1,i}, A_{1,i}, \ldots, L_{K,i}, Y_i),$$

where $L_t \in \mathcal{L}_t$ denotes time-t covariates, $A_t \in \mathcal{A}_t$ denotes the treatment/action at time t, and $Y \in \mathcal{Y}$ is a terminal outcome observed after time K. Define the history available immediately prior to the assignment of A_t as

$$H_t := (L_0, A_0, L_1, A_1, \ldots, L_t).$$

We assume i.i.d. sampling across units:

$$O_1, \ldots, O_n \overset{\text{i.i.d.}}{\sim} P_0.$$

All temporal dependence is contained within each trajectory.

Statistical model Let $\mathcal{M}$ denote a nonparametric statistical model for the distribution P_0 of O. We assume that P_0 admits a density with respect to a suitable product dominating measure and factorizes as

$$p_0(o) = p_0(l_0) \prod_{t=0}^{K} g_{t,0}(a_t \mid h_t) \prod_{t=1}^{K} p_0(l_t \mid h_{t-1}, a_{t-1})\, p_0(y \mid h_K, a_K), \tag{1}$$

where $g_{t,0}(\cdot \mid h_t)$ denotes the (unknown) logging policy at time t.

No parametric assumptions are imposed on the components of (1). In particular, the joint distribution of histories (H_K, A_K) is left completely unrestricted.

Target functional Let $\pi = (\pi_0, \ldots, \pi_K)$ denote a (possibly stochastic) intervention policy, where each $\pi_t(\cdot \mid h_t)$ is a probability distribution on $\mathcal{A}_t$.

Let $k_Y : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ be a measurable, bounded kernel with reproducing kernel Hilbert space (RKHS) $\mathcal{H}_Y$ and feature map $\varphi(y) := k_Y(y, \cdot)$.

For any distribution $P \in \mathcal{M}$, define P^π as the interventional distribution induced by replacing the treatment mechanisms $(g_t)_{t=0}^K$ in the factorization of P with $(\pi_t)_{t=0}^K$, leaving all other conditional laws unchanged.

The target functional is the longitudinal counterfactual mean embedding

$$\Psi_\pi(P) := \mathbb{E}_{P^\pi}\big[\varphi(Y)\big] \;\in\; \mathcal{H}_Y, \qquad P \in \mathcal{M}, \tag{2}$$

and the corresponding target parameter is

$$\mu^\pi := \Psi_\pi(P_0). \tag{3}$$

Static, dynamic, and stochastic treatment regimes are all encompassed within this formulation.

The primary nuisance components indexing μ^π are the nested conditional mean embeddings

$$\mu_{K+1}(h_K, a_K) := \mathbb{E}_{P_0}\big[\varphi(Y) \mid H_K = h_K, A_K = a_K\big], \tag{4}$$

together with their backward recursive extensions arising from the longitudinal structure.

Example 1.1 (Mediated response embedding)**.** *Consider (X, A, M, Y), where X are baseline covariates, A a treatment, M a post-treatment mediator, and $Y \in \mathcal{Y}$ a terminal outcome. This corresponds to the longitudinal structure with $K = 0$ by identifying $L_0 = X$, $A_0 = A$, and $L_1 = M$. Let $\pi = (\pi_A, \pi_M)$ be a stochastic intervention, where $\pi_A(\cdot \mid X)$ and $\pi_M(\cdot \mid X, A)$ specify interventions on A and M, respectively. The associated target is the mediated counterfactual mean embedding*

$$\mu^\pi = \mathbb{E}\big[\varphi\big(Y^{A\sim\pi_A,\, M\sim\pi_M}\big)\big] \;\in\; \mathcal{H}_Y,$$

which coincides with $\Psi_\pi(P_0)$ under the identification assumptions stated in the next section.

2 Nested representation

2.1 Identification assumptions

We impose the following conditions to relate the causal estimand of interest to the statistical target parameter $\mu^\pi = \Psi_\pi(P_0)$.

Assumption 2.1 (Consistency)**.** *For any treatment regime $\pi = (\pi_0, \ldots, \pi_K)$, the observed outcome satisfies $Y = Y^\pi$ almost surely whenever the realized treatment history $(A_0, \ldots, A_K)$ coincides with π.*

Assumption 2.1 links observed outcomes to their corresponding counterfactuals under the realized treatment path.

Assumption 2.2 (Sequential ignorability)**.** *For each $t = 0, \ldots, K$ and any treatment regime π,*

$$Y^\pi \;\perp\!\!\!\perp\; A_t \,\big|\, H_t \quad \text{under } P_0.$$

Assumption 2.2 rules out unmeasured time-varying confounding conditional on the observed history.

Assumption 2.3 (Positivity)**.** *For each $t = 0, \ldots, K$ and for P_0-almost every history h_t,*

$$\pi_t(a \mid h_t) > 0 \quad \Longrightarrow \quad g_{t,0}(a \mid h_t) > 0, \qquad \forall a \in \mathcal{A}_t.$$

Assumption 2.3 ensures that the target intervention is supported by the observed treatment assignment mechanism.

2.2 Nested g-computation

Under Assumptions 2.1–2.3, the target parameter μ^π is nonparametrically identified and admits a nested representation in $\mathcal{H}_Y$.

Proposition 2.4 (Identification via nested g-computation)**.** *Assume Assumptions 2.1–2.3. Let $\pi = (\pi_0, \ldots, \pi_K)$ be a (possibly stochastic) intervention policy. Let k_Y be a bounded kernel on $\mathcal{Y}$ with RKHS $\mathcal{H}_Y$ and feature map $\varphi(y) = k_Y(y, \cdot)$.*

Define recursively the sequence of measurable $\mathcal{H}_Y$-valued functions $(Q_t^\pi)_{t=0}^{K+1}$ by

$$Q_{K+1}^\pi(H_{K+1}) := \varphi(Y),$$

and, for each $t = 0, \dots, K$,

$$\bar{Q}^{\pi}_{t+1}(h_t, a_t) := \mathbb{E}_{P_0}\left[Q^{\pi}_{t+1}(H_{t+1}) \,\middle|\, H_t = h_t,\ A_t = a_t\right], \tag{5}$$

$$Q^{\pi}_t(h_t) := \int \bar{Q}^{\pi}_{t+1}(h_t, a)\, \pi_t(da \mid h_t), \tag{6}$$

where the conditional expectation is understood as a Bochner expectation in $\mathcal{H}_Y$ and the integral is finite in $\mathcal{H}_Y$. Then the target parameter satisfies

$$\mu^{\pi} = \Psi_{\pi}(P_0) = \mathbb{E}_{P_0}[Q^{\pi}_0(H_0)] \in \mathcal{H}_Y. \tag{7}$$

In particular, when $\mathcal{A}_t$ is discrete, (6) reduces to $Q^{\pi}_t(h_t) = \sum_{a \in \mathcal{A}_t} \bar{Q}^{\pi}_{t+1}(h_t, a)\, \pi_t(a \mid h_t)$. The sequence $(Q^{\pi}_t)_{t=0}^{K+1}$ is defined by backward recursion and can be interpreted as a sequence of $\mathcal{H}_Y$-valued value functions under π. Starting from $Q^{\pi}_{K+1} = \varphi(Y)$, each step applies: (i) a conditional (Bochner) expectation under P_0, propagating the embedding backward through the longitudinal structure, and (ii) integration with respect to π_t, replacing the observed treatment mechanism at time t. This alternating regression–integration structure respects the causal ordering and yields a nested functional of P_0 whose value at $t = 0$ equals the counterfactual mean embedding μ^{π}.

Remark. *The recursion (5)–(6) is well-defined under mild integrability conditions, e.g. $\mathbb{E}_{P_0}[\|\varphi(Y)\|_{\mathcal{H}_Y}] < \infty$ and $\bar{Q}^{\pi}_{t+1}(H_t, A_t) \in L^1(P_0; \mathcal{H}_Y)$ for all t.*

Proposition 2.4 characterizes μ^{π} as a well-defined functional of P_0 and reveals (i) a nested composition of conditional expectation operators, (ii) stagewise residuals $Q^{\pi}_{t+1}(H_{t+1}) - \bar{Q}^{\pi}_{t+1}(H_t, A_t)$ in $\mathcal{H}_Y$, and (iii) dependence on P_0 through iterated conditional mean embeddings, which will be central for efficient estimation.

Example 2.5 (Mediated response embedding (continued))**.** *Under Assumptions 2.1–2.3, the mediated counterfactual mean embedding under (π_A, π_M) satisfies*

$$\mu^{\pi} = \mathbb{E}_{P_0}[Q_0(X)] \ \in\ \mathcal{H}_Y, \tag{8}$$

where

$$Q_2(x, a, m) = \mathbb{E}_{P_0}[\varphi(Y) \mid X = x, A = a, M = m], \tag{9}$$

$$Q_1(x, a) = \int Q_2(x, a, m)\, \pi_M(dm \mid x, a), \tag{10}$$

$$Q_0(x) = \int Q_1(x, a)\, \pi_A(da \mid x). \tag{11}$$

This provides the simplest nontrivial instance of a longitudinal counterfactual mean embedding, already exhibiting multiple nested conditional mean embeddings.

3 Canonical gradient and stagewise Riesz representation

3.1 Canonical gradient

We characterize the pathwise derivative of the counterfactual mean embedding functional $\Psi_{\pi} : \mathcal{M} \to \mathcal{H}_Y$ and identify its Hilbert-space canonical gradient.

Let $\{P_{\epsilon} : \epsilon \in (-\delta, \delta)\} \subset \mathcal{M}$ be any regular parametric submodel through P_0, with score

$$s(O) = \frac{d}{d\epsilon} \log p_{\epsilon}(O)\bigg|_{\epsilon=0} \in L^2_0(P_0). \tag{12}$$

Theorem 3.1 (Pathwise differentiability and canonical gradient)**.** *Assume Assumptions 2.1–2.3 and $\mathbb{E}_{P_0} \|\varphi(Y)\|^2_{\mathcal{H}_Y} < \infty$. Then Ψ_{π} is pathwise differentiable at P_0 as a map into $\mathcal{H}_Y$.*

There exists a unique canonical gradient

$$\Phi_{\pi} \in L^2_0(P_0; \mathcal{H}_Y) \tag{13}$$

such that, for every regular submodel,

$$\frac{d}{d\epsilon}\Psi_\pi(P_\epsilon)\Big|_{\epsilon=0} = \mathbb{E}_{P_0}[\Phi_\pi(O)\, s(O)] \;\in\; \mathcal{H}_Y. \tag{14}$$

The canonical gradient admits the representation

$$\Phi_\pi(O) = \sum_{t=0}^{K} W_t^\pi(H_t, A_t)\Big(Q_{t+1}^\pi(H_{t+1}) - \bar{Q}_{t+1}^\pi(H_t, A_t)\Big) + Q_0^\pi(H_0) - \mu^\pi, \tag{15}$$

where

$$W_t^\pi(H_t, A_t) = \prod_{s=0}^{t} \frac{\pi_s(A_s \mid H_s)}{g_{s,0}(A_s \mid H_s)}. \tag{16}$$

Theorem 3.1 reveals the structure of the Hilbert-valued canonical gradient. It preserves the backward recursion of nested g-computation and decomposes into stagewise $\mathcal{H}_Y$-valued residuals $Q_{t+1}^\pi(H_{t+1}) - \bar{Q}_{t+1}^\pi(H_t, A_t)$ weighted by W_t^π. The functional is linear in $\varphi(Y)$ and depends on P_0 only through iterated conditional mean embeddings, ensuring orthogonality in $L^2(P_0; \mathcal{H}_Y)$. Because the gradient is a sum of weighted residual operators, errors propagate forward in time, and each stage induces a distinct inverse problem, motivating the stagewise Riesz representation.

3.2 Stagewise Riesz representation

To make the linear structure explicit, define for each stage $t = 0, \ldots, K$ the conditional expectation operator

$$(\mathcal{T}_t f)(h_t, a_t) := \mathbb{E}_{P_0}[f(H_{t+1}) \mid H_t = h_t,\; A_t = a_t]\,, \tag{17}$$

and the associated residual operator

$$(\mathcal{R}_t f)(O) := f(H_{t+1}) - (\mathcal{T}_t f)(H_t, A_t). \tag{18}$$

With this notation, each summand in the canonical gradient in Theorem 3.1 can be written as

$$W_t^\pi(H_t, A_t)\,(\mathcal{R}_t Q_{t+1}^\pi)(O), \tag{19}$$

that is, as a weighted residual of $\mathcal{T}_t$ applied to the $\mathcal{H}_Y$-valued function Q_{t+1}^π.

Stagewise Riesz functionals For each t, define the bounded linear functional ℓ_t^π acting on scalar functions f on the space of histories H_{t+1} by

$$\ell_t^\pi(f) := \mathbb{E}_{P_0}[W_t^\pi(H_t, A_t)\,(\mathcal{R}_t f)(O)]\,. \tag{20}$$

In the embedding setting, the target parameter admits the representation

$$\mu^\pi = \sum_{t=0}^{K} \mathbb{E}_{P_0}\left[W_t^\pi(H_t, A_t)\,(\mathcal{R}_t Q_{t+1}^\pi)(O)\right], \tag{21}$$

so that each stage contributes through the action of ℓ_t^π applied coordinate-wise to the $\mathcal{H}_Y$-valued function Q_{t+1}^π.

Let $\mathcal{H}_{t+1}$ be a Hilbert space of scalar functions on the domain of H_{t+1} such that ℓ_t^π is continuous on $\mathcal{H}_{t+1}$. By the Riesz representation theorem, for each t there exists a unique scalar function

$$r_t^\pi \in \mathcal{H}_{t+1} \tag{22}$$

satisfying

$$\ell_t^\pi(f) = \langle r_t^\pi, f\rangle_{\mathcal{H}_{t+1}}, \qquad \forall f \in \mathcal{H}_{t+1}. \tag{23}$$

Definition 3.2 (Stagewise Riesz representers)**.** The functions $(r_t^\pi)_{t=0}^K$ are called the *stagewise Riesz representers* associated with the counterfactual mean embedding functional Ψ_π.

The longitudinal structure therefore induces not a single representer, but a sequence of stagewise scalar representers, each acting on its own history space. The target embedding decomposes as

$$\mu^\pi = \sum_{t=0}^{K} \mathbb{E}_{P_0}\left[r_t^\pi(H_{t+1})\, Q_{t+1}^\pi(H_{t+1})\right]. \tag{24}$$

Estimating $(r_t^\pi)_{t=0}^K$ jointly would yield a single coupled inverse problem over $\mathcal{H}_1 \oplus \cdots \oplus \mathcal{H}_{K+1}$. In contrast, the decomposition above leads naturally to $K+1$ decoupled Riesz regressions, each defined on its own history space and admitting stage-specific regularization.

4 Estimation

We now describe an estimator of the counterfactual mean embedding μ^π based on the stagewise Riesz representation derived in Section 3. Given estimators $(\widehat{r}_t^\pi, \widehat{Q}_{t+1}^\pi)_{t=0}^K$, we define the one-step embedding estimator

$$\widehat{\mu}^\pi := \sum_{t=0}^{K} \frac{1}{n} \sum_{i=1}^{n} \widehat{r}_t^\pi(H_{t+1,i})\, \widehat{Q}_{t+1}^\pi(H_{t+1,i}) \;\in\; \mathcal{H}_Y. \tag{25}$$

The remainder of this section specifies the construction of $\widehat{r}_t^\pi$ and $\widehat{Q}_{t+1}^\pi$.

4.1 Stagewise Riesz regression

For each stage t, recall the bounded linear functional

$$\ell_t^\pi(f) = \mathbb{E}_{P_0}[W_t^\pi(H_t, A_t)\,(\mathcal{R}_t f)(O)]. \tag{26}$$

Let $\mathcal{H}_{t+1}$ be a Hilbert space of scalar functions on the domain of H_{t+1}. The stagewise Riesz representer $r_t^\pi \in \mathcal{H}_{t+1}$ is the unique solution to the population normal equation

$$\mathbb{E}_{P_0}[r_t^\pi(H_{t+1})\, f(H_{t+1})] = \ell_t^\pi(f), \qquad \forall f \in \mathcal{H}_{t+1}. \tag{27}$$

In practice, we approximate (27) by solving the regularized empirical problem

$$\widehat{r}_t^\pi = \arg\min_{r \in \mathcal{H}_{t+1}} \left\{ \frac{1}{n} \sum_{i=1}^{n} \Big(r(H_{t+1,i}) - \widehat{\zeta}_{t,i} \Big)^2 + \lambda_t \|r\|_{\mathcal{H}_{t+1}}^2 \right\}, \tag{28}$$

where

$$\widehat{\zeta}_{t,i} := W_t^\pi(H_{t,i}, A_{t,i}) - \widehat{\mathcal{T}}_t^* W_t^\pi(H_{t,i}, A_{t,i}) \tag{29}$$

denotes the empirical moment implied by (26), and $\widehat{\mathcal{T}}_t^*$ is an estimator of the conditional expectation operator.

Stagewise versus global regression. A global approach would estimate all representers jointly by solving a single Riesz regression over the product space $\bigoplus_{t=0}^K \mathcal{H}_{t+1}$. Such a procedure couples stages, imposes global regularization, and aggregates the ill-posedness of the conditional expectation operators across time.

In contrast, the stagewise formulation yields $K+1$ decoupled inverse problems aligned with the temporal structure of the canonical gradient. This separation permits stage-specific regularization and prevents amplification of instability across time.

4.2 Estimation of nested conditional mean embeddings

The nested conditional mean embeddings $(Q_{t+1}^\pi)_{t=0}^K$ are estimated via a backward recursive procedure analogous to g-computation.

Initialize

$$\widehat{Q}_{K+1}^\pi(H_{K+1}) := \varphi(Y) \;\in\; \mathcal{H}_Y. \tag{30}$$

For $t = K, \ldots, 0$, define

$$\widehat{\widehat{Q}}^{\pi}_{t+1}(h_t, a_t) := \widehat{\mathbb{E}}\Big[\widehat{Q}^{\pi}_{t+1}(H_{t+1}) \mid H_t = h_t,\ A_t = a_t\Big], \tag{31}$$

$$\widehat{Q}^{\pi}_{t}(h_t) = \int \widehat{\widehat{Q}}^{\pi}_{t+1}(h_t, a)\, \pi_t(da \mid h_t), \tag{32}$$

where the conditional expectations are estimated as $\mathcal{H}_Y$-valued regressions (i.e., conditional mean embedding estimators) using any suitable method.

Cross-fitting may be employed to mitigate overfitting and to ensure orthogonality between the estimation errors of $\widehat{Q}^{\pi}_{t+1}$ and those of the stagewise Riesz regressions.

Together, the estimators $(\widehat{r}^{\pi}_t, \widehat{Q}^{\pi}_{t+1})_{t=0}^{K}$ define the stagewise embedding estimator $\widehat{\mu}^{\pi}$ in (25), whose statistical properties are analyzed in the next section.

4.3 Consistency

Theorem 4.1 (Consistency and rate for nested counterfactual mean embedding)**.** *Fix π. Under Assumptions 2.1–2.3, $\mathbb{E}\,\|\varphi(Y)\|^2_{\mathcal{H}_Y} < \infty$, and cross-fitting, the one-step estimator $\widehat{\mu}^{\pi}$ admits the expansion*

$$\widehat{\mu}^{\pi} - \mu^{\pi} = \frac{1}{n}\sum_{i=1}^{n} \Phi_{\pi}(O_i) + \mathrm{Rem}_n, \tag{33}$$

where $\Phi_{\pi} \in L^2_0(P_0; \mathcal{H}_Y)$ is the canonical gradient in Theorem 3.1 and the remainder satisfies

$$\|\,\mathrm{Rem}_n\,\|_{\mathcal{H}_Y} \lesssim \sum_{t=0}^{K} \|\widehat{r}^{\pi}_t - r^{\pi}_t\|_{L^2(P_0)}\, \|\widehat{Q}^{\pi}_{t+1} - Q^{\pi}_{t+1}\|_{L^2(P_0;\mathcal{H}_Y)} + o_{\mathbb{P}}(n^{-1/2}). \tag{34}$$

Consequently, if for every t,

$$\|\widehat{r}^{\pi}_t - r^{\pi}_t\|_{L^2(P_0)} = o_{\mathbb{P}}(1), \qquad \|\widehat{Q}^{\pi}_{t+1} - Q^{\pi}_{t+1}\|_{L^2(P_0;\mathcal{H}_Y)} = o_{\mathbb{P}}(1), \tag{35}$$

then $\|\widehat{\mu}^{\pi} - \mu^{\pi}\|_{\mathcal{H}_Y} = o_{\mathbb{P}}(1)$. Moreover, if for every t,

$$\|\widehat{r}^{\pi}_t - r^{\pi}_t\|_{L^2(P_0)}\, \|\widehat{Q}^{\pi}_{t+1} - Q^{\pi}_{t+1}\|_{L^2(P_0;\mathcal{H}_Y)} = o_{\mathbb{P}}(n^{-1/2}), \tag{36}$$

then $\|\widehat{\mu}^{\pi} - \mu^{\pi}\|_{\mathcal{H}_Y} = O_{\mathbb{P}}(n^{-1/2})$. In particular, if uniformly in t,

$$\|\widehat{r}^{\pi}_t - r^{\pi}_t\|_{L^2(P_0)} = O_{\mathbb{P}}(n^{-1/4}), \qquad \|\widehat{Q}^{\pi}_{t+1} - Q^{\pi}_{t+1}\|_{L^2(P_0;\mathcal{H}_Y)} = O_{\mathbb{P}}(n^{-1/4}), \tag{37}$$

then $\|\widehat{\mu}^{\pi} - \mu^{\pi}\|_{\mathcal{H}_Y} = O_{\mathbb{P}}(n^{-1/2})$.

5 Distributional effects

5.1 Q1

Fix two policies π and π'. We test

$$H_0 : \mu^{\pi} = \mu^{\pi'} \qquad \text{vs.} \qquad H_1 : \mu^{\pi} \neq \mu^{\pi'}. \tag{38}$$

Since k_Y is characteristic,

$$\mu^{\pi} - \mu^{\pi'} = \mathbb{E}_{P_0}[\Phi_{\pi,\pi'}(O)]\,, \tag{39}$$

where

$$\Phi_{\pi,\pi'}(O) := \Phi_{\pi}(O) - \Phi_{\pi'}(O), \tag{40}$$

and Φ_{π} denotes the nested longitudinal canonical gradient from Theorem 3.1.

Thus the null hypothesis is equivalently

$$H_0 : \mathbb{E}_{P_0}[\Phi_{\pi,\pi'}(O)] = 0. \tag{41}$$

Let $\Delta_{\pi,\pi'} := \mu^{\pi} - \mu^{\pi'} \in \mathcal{H}_Y$ and let $\widehat{\Phi}_{\pi,\pi'}(O)$ be the cross-fitted plug-in estimator of $\Phi_{\pi,\pi'}(O) = \Phi_{\pi}(O) - \Phi_{\pi'}(O)$. Split the sample into two disjoint sets $\mathcal{I}_1 = \{1, \ldots, m\}$ and $\mathcal{I}_2 = \{m+1, \ldots, n\}$ with $m = \lfloor n/2 \rfloor$. Fit all nuisance components on $\mathcal{I}_1$ and evaluate $\widehat{\Phi}_{\pi,\pi'}$ on $\mathcal{I}_2$ (and symmetrize by swapping the roles of the splits if desired).

For each $i \in \mathcal{I}_1$, define the cross inner-product statistic

$$f_i := \frac{1}{|\mathcal{I}_2|} \sum_{j \in \mathcal{I}_2} \left\langle \widehat{\Phi}_{\pi,\pi'}(O_i), \widehat{\Phi}_{\pi,\pi'}(O_j) \right\rangle_{\mathcal{H}_Y}. \tag{42}$$

Let

$$\bar{f} = \frac{1}{|\mathcal{I}_1|} \sum_{i \in \mathcal{I}_1} f_i, \qquad S^2 = \frac{1}{|\mathcal{I}_1|} \sum_{i \in \mathcal{I}_1} \left(f_i - \bar{f}\right)^2. \tag{43}$$

We consider the studentized test statistic

$$T_n^{\dagger} := \frac{\sqrt{|\mathcal{I}_1|}\, \bar{f}}{S}. \tag{44}$$

Theorem 5.1 (Asymptotic normality of the studentized statistic)**.** *Assume Assumptions 2.1–2.3 and* $\mathbb{E}_{P_0} \|\Phi_{\pi,\pi'}(O)\|^4_{\mathcal{H}_Y} < \infty$. *Assume cross-fitting and the rate condition*

$$\sum_{t=0}^{K} \|\widehat{r}_t - r_t\|_{L^2(P_0)} \, \|\widehat{Q}_{t+1} - Q_{t+1}\|_{L^2(P_0;\mathcal{H}_Y)} = o_{\mathbb{P}}(n^{-1/2}). \tag{45}$$

If $H_0 : \Delta_{\pi,\pi'} = 0$ *holds and the non-degeneracy condition*

$$\mathrm{Var}\Big(\langle \Phi_{\pi,\pi'}(O), \Phi_{\pi,\pi'}(O') \rangle_{\mathcal{H}_Y} \Big) > 0 \tag{46}$$

holds for $O, O' \overset{\text{i.i.d.}}{\sim} P_0$, *then*

$$T_n^{\dagger} \Rightarrow \mathcal{N}(0,1). \tag{47}$$

5.2 Q2